



Shijia Xu

M.S. Candidate in Control Science and Engineering

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🎓 Education

- **Queen Mary University of London** *London, United Kingdom*
Visiting Research Student in Computer Science *Jun. 2026 – Dec. 2026*
Host Supervisor: Prof. Ahmed M. A. Sayed (*Machine Learning Systems and Efficient Inference*)
Funding: Supported by the Chongqing University Joint Training Program for Master's Students
Research Focus: NLP and RAG for reliable and reasoning-capable Large Language Models
- **Chongqing University** *Chongqing, China*
M.E. Candidate in Control Science and Engineering; GPA: 4.09/5.00 (Rank: 2/74) *Sep. 2024 – Jul. 2027*
Advisor: Prof. Zhou Wu (*Optimization, Artificial Intelligence and Energy Systems*)
Research Interests: Trustworthy Large Language Models, Retrieval-Augmented Generation, and Explainable NLP
Selected Coursework: Graph Theory and Applications, Machine Learning and Big Data Processing, and Data Mining
- **Shandong University of Science and Technology** *Qingdao, China*
B.E. in Automation; GPA: 4.03/5.00 (Rank: 3/163) *Sep. 2020 – Jul. 2024*
Selected Coursework: Principles of Automatic Control, Intelligent Control, and Advanced Mathematics
Selected Projects: ESN-Based Wind Energy Forecasting; FPGA-Based Wind Speed Measurement

📖 Publication

- **Self-Correcting RAG: Enhancing Faithfulness via MMKP Context Selection and NLI-Guided MCTS**
Shijia Xu, Zhou Wu, Xiaolong Jia, Yu Wang, Kai Liu, April Xiaowen Dong. *ACL 2026*
 - Introduces a unified RAG framework that formulates context selection as a Multi-dimensional Multiple-choice Knapsack Problem under a strict token budget and employs NLI-guided Monte Carlo Tree Search to explore and verify reasoning trajectories, improving reasoning accuracy while reducing hallucinations.
- **RCBSF: A Multi-Agent Framework for Automated Contract Revision via Stackelberg Game**
Shijia Xu, Yu Wang, Xiaolong Jia, Zhou Wu, Kai Liu, April Xiaowen Dong. *ACL 2026*
 - Introduces a risk-constrained bilevel Stackelberg framework for automated contract revision, in which a global prescriptive agent imposes explicit risk budgets on revision and verification agents, enabling risk-aware optimization with theoretical convergence guarantees.
- **LLM-Guided Secure Federated Visual Prompts with Deep Unfolding for MRI Reconstruction**
Di Xiao, Yuhan Gou, Yu Ren, Shijia Xu, Yue Zhang. *ICMR 2026*
 - Introduces FedLUR, a secure federated MRI reconstruction framework that combines deep unfolding, lightweight visual prompts, and privacy-preserving LLM-guided aggregation.
- **State Copying Crowds Out Reasoning: Mechanistic Evidence for Delta Planning in Autoregressive Models**
Wang Xi, Shijia Xu* (co-first author).* *NeurIPS 2026 (In Submission)*
 - Identifies copy-reason interference in autoregressive planning, showing that full-state generation disrupts goal-directed computation. Demonstrates through controlled experiments and activation patching that delta-style planning substantially improves long-horizon reasoning.

- **Resource-Aware Federated LoRA Fine-Tuning for Heterogeneous Environments**

Xiaolong Jia, Ahmed M. Abdelmoniem, Shijia Xu, Gaoyang Liu, Chen Wang. *NeurIPS 2026 (In Submission)*

- Proposes resource-aware federated fine-tuning with budget-adaptive LoRA ranks, Top-*k* sparsification, and selective regularization for efficient communication and robust adaptation.

- **The Cost of Compression: A Rate-Distortion Limit on Factual Hallucination**

Wang Xi*, Shijia Xu* (co-first author), Rongfeng Guo. *AACL 2026 (In Submission)*

- Establishes an information-theoretic coverage-compression framework that decomposes factual hallucination into compression distortion on observed facts and missing coverage on unseen facts.

- **MemSIF: From Structured Interactions to Dual-Track Fact Memory for LLM Agents**

Yufei Luo, Shijia Xu, Guangyuan Dong, Xiucheng Xu. *AAAI 2027 (In Submission)*

- Introduces MemSIF, combining structured interaction memory with CoreFact and ActiveFact tracks to improve long-term agent memory on LoCoMo and LongMemEval-S.

Project Experience

- **Intelligent Contract Management System**

Core Developer and Project Lead *Jan. 2025 – Oct. 2025*

- Developed a full-stack contract management platform using Flask and Redis, featuring role-based access control, real-time communication, document versioning, and template management. Integrated DeepSeek LLM for automated contract generation, legal risk analysis, and clause extraction.

- **Robot Team, Shandong University of Science and Technology**

Team Leader and Head of the Control & Electronics Group *Dec. 2021 – May 2023*

- Led a 30-member team in the ABU Robocon competition, coordinating robot development and competition strategy. Oversaw embedded control programming, hardware design, sensor integration, and system-level debugging for manual and autonomous robots.

Honors

- [1] Chongqing University First Prize Master's Academic Scholarship (2024 & 2025)
- [2] Chongqing University Merit Student (2025)
- [3] Outstanding Graduate of Shandong Province (2024)
- [4] Shandong Provincial Government Scholarship (2022)
- [5] Sun Yueqi Outstanding Student Award (*Top Industry Honor*) (2022)
- [6] National Endeavor Scholarship (*Top 3%*) (2021)

Awards

- [1] 15th MathorCup Math Application Challenge (Postgraduate) - National Second Prize (2025)
- [2] RAICOM Developer Robot Competition - National Second Prize (2023)
- [3] Nat'l Undergrad Robotics Competition (ROBOCON) - National Third Prize (2022 & 2023)
- [4] Int'l Advanced Robot & Simulation Competition - National Second Prize (2022)
- [5] MathorCup Math Modeling Challenge - National Third Prize (2022)
- [6] Intelligent Control Contest - Provincial First Prize (2022)
- [7] Nat'l Undergrad Electronics Design Contests - Provincial Second Prize (2021 & 2022)

Professional Skills

- **Programming Skills:** Proficient in Python, Java, SQL, Excel, PowerPoint; Experienced in PyTorch and TensorFlow machine learning development; Intermediate in MATLAB, C; Relevant projects in NLP and CV.
- **Interests:** Texas Hold'em Poker, travel blogging, photography, and basketball.